



Virtual Reality: The Next Disruptor

A new kind of worldwide communication.

By Philip Rosedale

VIRTUAL REALITY (VR) WILL BE A PROFOUNDLY DISRUPTIVE technology in the same way as the smartphone and the Internet. Like the smartphone, VR uses a new interface format (a head-mounted display and hand controllers) to provide much more intuitive and natural access to a computing device. Much like the Internet, it allows a new kind of worldwide communication—but this time with a natural human experience that will be nearly indistinguishable from standing face to face or in a group. The combination of these two capabilities, along with a price point similar to the first smartphones, suggests that we will see an adoption curve similar to that of the smartphone, with a majority of worldwide Internet users making daily use of VR in the next seven to ten years.

VR technology has been demonstrated multiple times in prior years, in ways that were nearly as compelling as the currently available Oculus Rift, HTC Vive, and Samsung GearVR devices. The big difference this time has been driven by the intense competition in the smartphone market to reduce the size and cost of orientation sensors and high-resolution displays to finally make these research devices available for consumers. The same advances have made lightweight wireless handheld controllers that track the motion of the hands and/or fingers with very high accuracy

possible. Another factor that has changed is Internet speed and latency. One-way Internet transmission times, even half-way around the world, are now short enough to allow two people to talk face to face as digital avatars without being able to detect the delay, making communication completely natural (Figure 1). This level of interaction had to wait for Internet speeds and routers to reach their current levels of performance, which are now close to the limits set by the speed of light.

An important initial observation about VR and augmented reality (AR) is that the rise of these devices will not depend on any kind of a “killer app” to drive adoption, because of the simple fact that they will replace desktop (or mobile) displays for many of the things we are using them for today. Once head-mounted displays (HMDs) can offer the same pixel resolution (in terms of pixels-per-angular-degree) as desktop or laptop displays, we will be able to paint our browsers or e-mail clients onto a huge surface surrounding us such as the walls of the conference room we are sitting in or the seat back of the airline chair in front of us. Turning all of the space around your head into a display is like having 100 standard desktop monitors that follow you everywhere. This is a huge improvement over the traditional monitor, and our appetite for visual information is limitless, as demonstrated by our passion for ever-larger screens and multimonitor displays. Currently, VR displays have about a quarter of the needed angular resolution to match the desktop display, but mounting the display on the head (as opposed to the



desktop) means that the constant small motions of the head give us a free form of anti-aliasing that makes the effective resolution of the screens higher. Many are surprised to discover how easily we can read small text or simulated desktop surfaces even in first-generation headsets. We can expect that second-generation VR headsets will begin to replace desktop displays for at least some applications in the next couple of years, regardless of which new native VR applications catch on.

Although entertainment and 360° video are capturing much of the early conversation around VR applications, in the long term, VR will make its greatest impact by enabling the creation of virtual worlds that we can visit together with others (Figure 2). These worlds will allow us to work, play, and meet each other in spaces that will be incredibly lifelike but will also offer capabilities far beyond the real world. As second-generation VR headsets capture eye movements and facial expressions, the experience of having a face-to-face business meeting will become possible simply by putting on an HMD rather than getting on an airplane. With business travel accounting for some 30% of global energy use, replac-

ing even a fraction of this amount would be a huge benefit to humanity as well as a very disruptive change (Figure 1).

Education will be another big early use case for virtual worlds. Imagine kids being able to attend class or tutoring by putting on an HMD and being immersed in interactive educational materials on subjects such as physics or chemistry, while being led and questioned by their teacher exactly as if they were seeing them in a real classroom (Figure 3). Although first-generation HMDs aren't cheap, the value to a student of distance learning using one will be easily worth the investment.

Pre-HMD virtual worlds such as Second Life demonstrate some of the promise that we are likely to see in HMD-enabled virtual worlds. Although users of Second Life have to go through the very steep learning curve of entering and manipulating the 3-D world using only a mouse and keyboard, the million or so regular users there generate close to a billion dollars a year in financial transactions and use the platform for a variety of purposes, including training, online education, historical recreation, support groups, and art galleries (Figure 4).



FIGURE 1. A business workshop conducted using HMDs and hand controllers.



FIGURE 5. Users sharing creations with one another.



FIGURE 2. Using HMDs and hand controllers to interact in an environment.



FIGURE 3. Changing how we teach.



FIGURE 4. Large groups of people from around the world gathering in a shared space.

It seems likely that with the rise of HMDs, we may see the use and scale of these general purpose virtual worlds expand greatly as the users become mainstream, and looking closely at what has already happened in them may give us clues as to how virtual worlds are likely to develop.

For virtual worlds to become successful, software and standards must be created, allowing many different people and companies to put up spaces in a manner similar to how the early Internet grew as a connected web of sites. We will need to standardize forms of identity, for example, so that we can appear on different servers with the same desired personal appearance. Digital content such as furniture, teaching materials, and architectural components with different types of interactive capability will need to be portable from one server to another (Figure 5). Beyond the HMD, software must be written to allow live 3-D audio, networking for synchronizing changes made to virtual environments, and approaches for scaling to thousands or even millions of concurrent users in the same locations. Standards for the protocols connecting VR clients to VR servers also need to be built, debated, maintained, and evolved. These are all things that we are working on at High Fidelity. There are no fundamentally unsolvable problems here. The existence of fast networks and HMDs makes the emergence of software that allows people to gather in virtual worlds of their own creation as inevitable as the rapid growth of websites in the late 1990s. It will be a very exciting time to be alive as we, for the first time as human beings, create new worlds for ourselves to inhabit.

ABOUT THE AUTHOR

Philip Rosedale (philip@highfidelity.io) is the chief executive officer and cofounder of High Fidelity, Inc., a company devoted to exploring the future of next-generation shared virtual reality. Prior to High Fidelity, he created the virtual civilization Second Life, populated by 1 million active users generating US\$700 million in annual transaction volumes. In addition to numerous technology inventions (including the video conferencing product FreeVue, which was acquired by RealNetworks in 1996, where he later served as chief technology officer), he has also worked on experiments in distributed work and computing.

